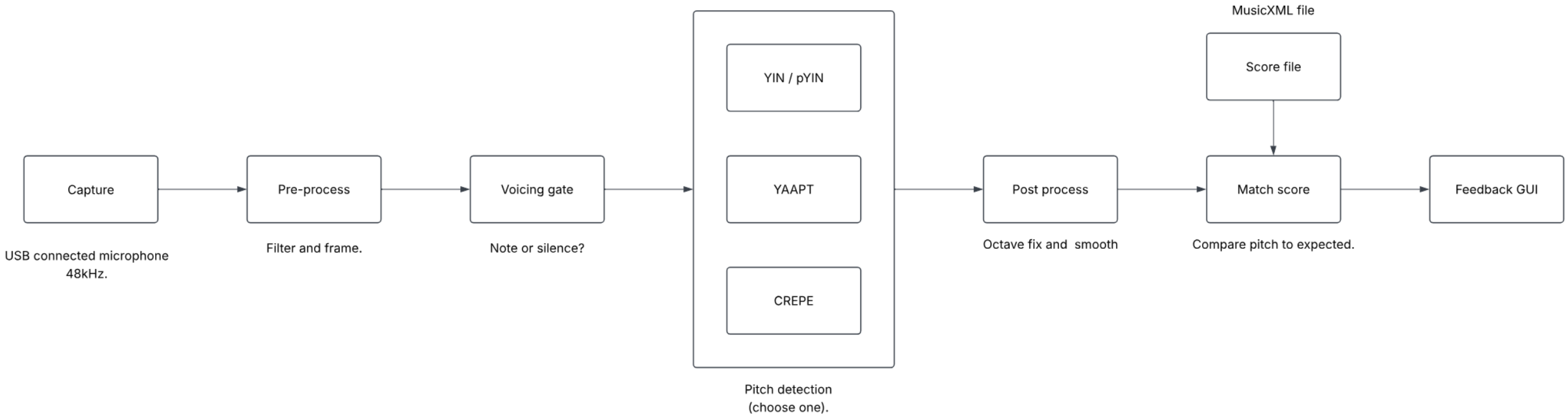
MREN 403 Project Proposal Meeting

Subsystem Review

- Audio Processing
- Video Processing
- Display / User Interface

Audio Processing (Flowchart)



Audio Processing (Microphone)

- Frequency
  - 48kHz microphone. Standard smartphone sample rate.
  - 96kHz microphone. Only use if the 48kHz is insufficient.
- Location
  - Stand mounted.
    - Easy to setup.
    - Low cost.
  - Cello mounted.
    - Higher sound quality.
    - Wired (cable management).
    - Wireless (high cost, wireless delay).
- Max SPL
  - 120dB or more. Cello reaches 100dB at close range.

Video Processing

- Google MediaPipe
  - 6 landmarks for 1 arm, 21 additional in the hand
- YOLOv8
  - Custom Ultralytics YOLOv8-pose model
  - Used for Cello landmarks (bow, bridge, fingerboard)
  - Our own data and annotation
- Use vectors from landmarks to find angles in arm, elbow, wrist, bow, and bridge
- Recognize rotation, pronation and make decision on posture corrections

Display / User Interface

- Tablet
  - Best case
  - Large screen for user, large camera FOV to capture player and cello
  - Samsung Galaxy Tab – 120 deg FOV
  - Modern iPads – 122 deg FOV
  - Modern laptops – 60-90 deg
- Laptop
  - Easiest for our purposes
  - USB webcam could help solve FOV